

Developer Technology Trends: Current Usage and Future Demand

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Python

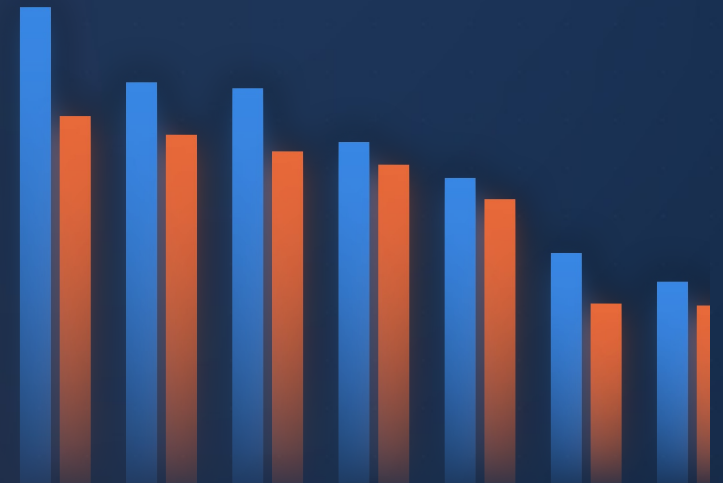
pandas

SQL / SQLite

Matplotlib

Seaborn

Looker Studio



From 65,437 raw survey responses to clear hiring signals

65,437

survey responses after removing 10 duplicate rows

18,845

respondents in the dashboard long table

161

countries represented in the survey

3

data sources: Job API, web scraping and the Stack Overflow Developer Survey

978

compensation outliers detected with the IQR method

64%

missing compensation values, imputed with the median (\$65,000)

What this deck covers

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Appendix

A stable core stack, with clear risers for next year

Key findings from job postings, salary data and 18,845 developer survey responses



Jobs & pay

Washington DC (5,316), Detroit (3,945) and Seattle (3,375) lead job postings; Swift (\$130.8K) and Python (\$114.4K) top the salary ranking



Languages

JavaScript, SQL and TypeScript are both the most used and the most desired languages. Rust and Go enter the desired top 10, while PHP and PowerShell drop out



Databases

PostgreSQL is the leading database today (11,514) and its lead grows next year (12,193); Redis rises to 2nd place



Platforms

AWS, Microsoft Azure and Google Cloud dominate cloud platforms; React and Node.js dominate web frameworks



Respondents

Most respondents are 25-44 years old (68.6%) and hold a bachelor's or master's degree (72.3%)

Which skills are in demand today, and which come next?

Purpose of the report, target audience & value of the analysis



Purpose

Identify which technology skills are in demand today and which ones developers want to adopt next



Target audience

Hiring managers, HR and training teams, and job seekers planning their skills



Value

Data-driven guidance on which languages, databases, platforms and frameworks to learn or invest in

Scope of the analysis

- Job postings by US city and salaries by programming language
- Developer survey: current vs. desired technologies and respondent demographics

From three raw sources to one clean, analysis-ready dataset

Data sources, data collection & data wrangling steps



STEP 1

Data sources

- Job postings via a Job API (job-postings.xlsx)
- Salaries by language via web scraping (popular-languages.csv)
- Stack Overflow Developer Survey



STEP 2

Data wrangling in Python (pandas)

- Removed 10 duplicate rows (65,447 to 65,437 responses)
- Imputed missing categorical values with the mode: RemoteWork (10,631 rows, "Hybrid") and EdLevel (4,653 rows, "Bachelor's degree")
- Imputed missing ConvertedCompYearly (64%) with the median (\$65,000), then normalized it with Min-Max and Z-score scaling
- Detected 978 compensation outliers with the IQR method (above \$220,861)



STEP 3

Exploratory analysis

- Distributions, outliers and correlations (pandas, Matplotlib, Seaborn, SQL on SQLite)



STEP 4

Dashboard

- Multi-select answers split into a long table (18,845 respondents), null values excluded, built in Google Looker Studio

RESULTS

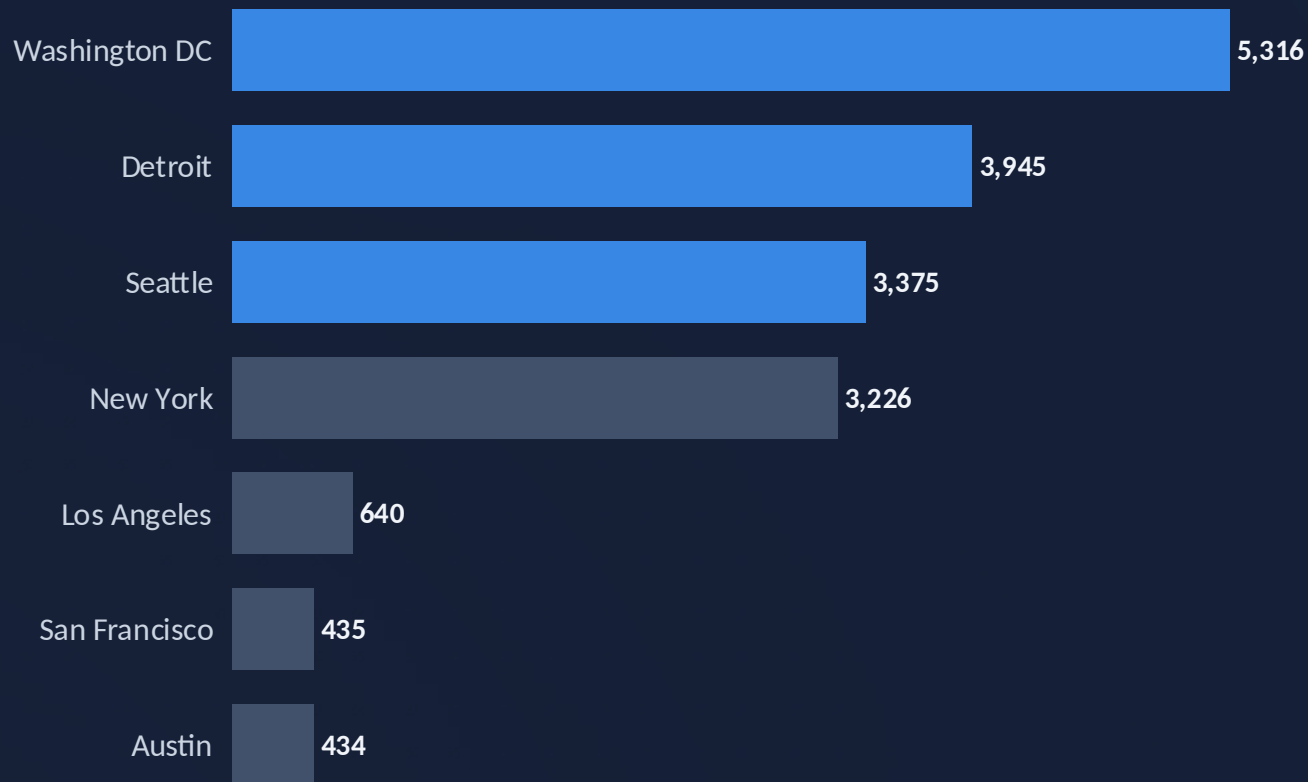
Where the jobs are, what pays, and what developers use

Job postings, salaries by language, and programming language and database trends

01

Washington DC leads job postings, ahead of Detroit and Seattle

Number of job postings by city



5,316

postings in Washington DC

Key takeaways

- Washington DC has the most postings (5,316), ahead of Detroit (3,945) and Seattle (3,375)
- New York follows closely (3,226)
- Los Angeles, San Francisco and Austin have far fewer postings (434-640)

Swift is the best-paid language; SQL and PHP pay the least

Average annual salary by programming language



\$130,801

average salary for Swift

Key takeaways

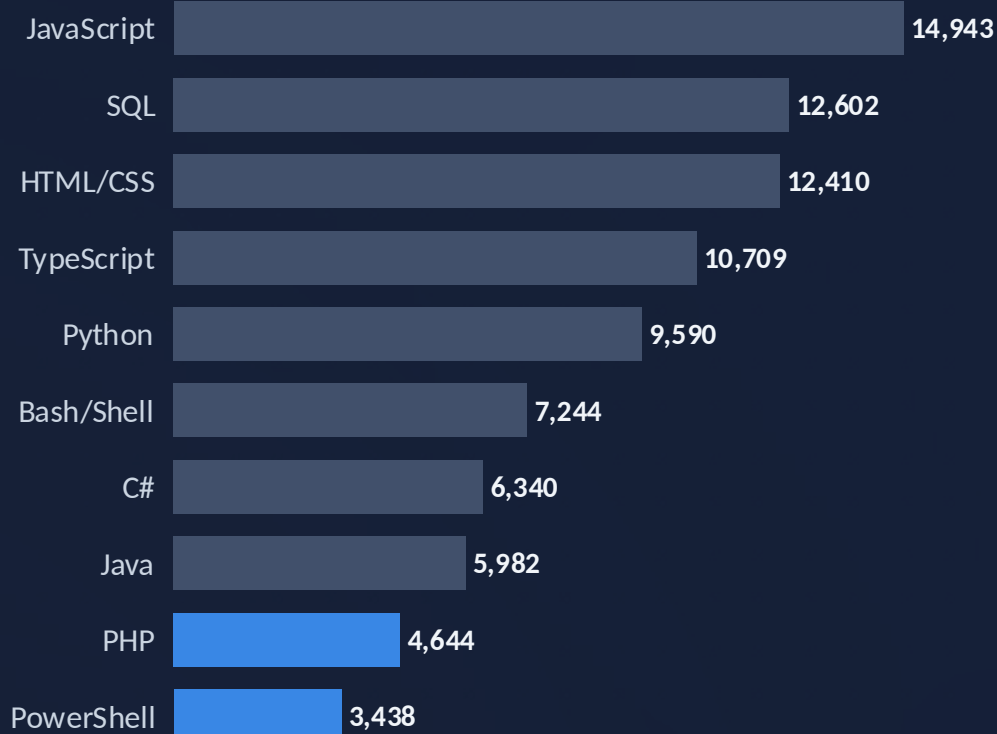
- Swift is the best-paid language (\$130,801)
- Python, C++ and JavaScript all exceed \$110,000
- SQL and PHP are at the bottom (~\$84,700-\$84,800), although SQL is one of the most used languages

Rust and Go enter next year's top 10; PHP and PowerShell drop out

JavaScript, SQL and TypeScript lead both years; Rust and Go replace PHP and PowerShell in next year's top 10.

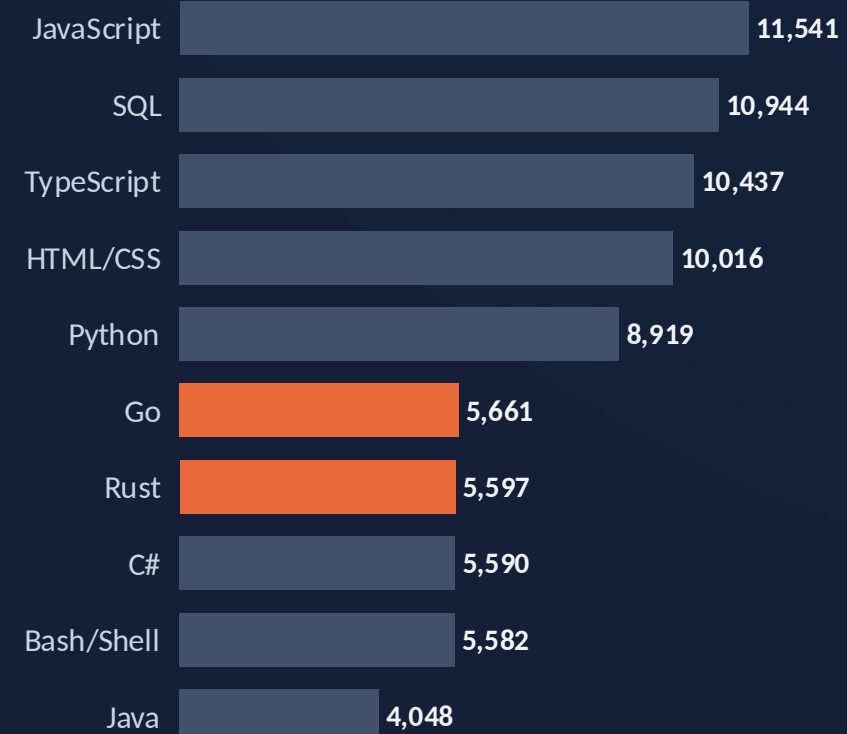
Current year: languages used

● Drops out of next year's top 10



Next year: languages desired

● New in next year's top 10



Web and data skills stay core; Rust and Go are the ones to watch

FINDINGS

- JavaScript is #1 in both years (14,943 used, 11,541 desired)
- TypeScript moves up from 4th to 3rd place, overtaking HTML/CSS
- Rust (5,597) and Go (5,661) enter the desired top 10; PHP and PowerShell leave it
- Python stays 5th in both years

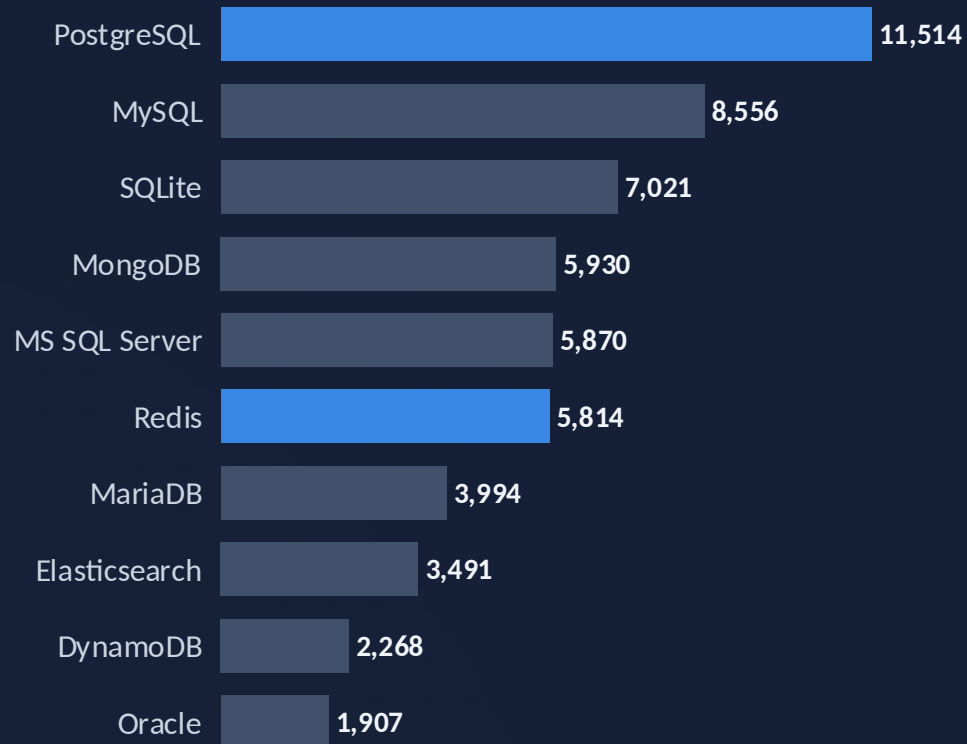
IMPLICATIONS

- Web skills (JavaScript/TypeScript) remain a safe core investment
- SQL stays essential for any data-related role
- Training in Rust and Go can anticipate growing demand for systems and cloud development
- Legacy languages (PHP, PowerShell) are losing interest

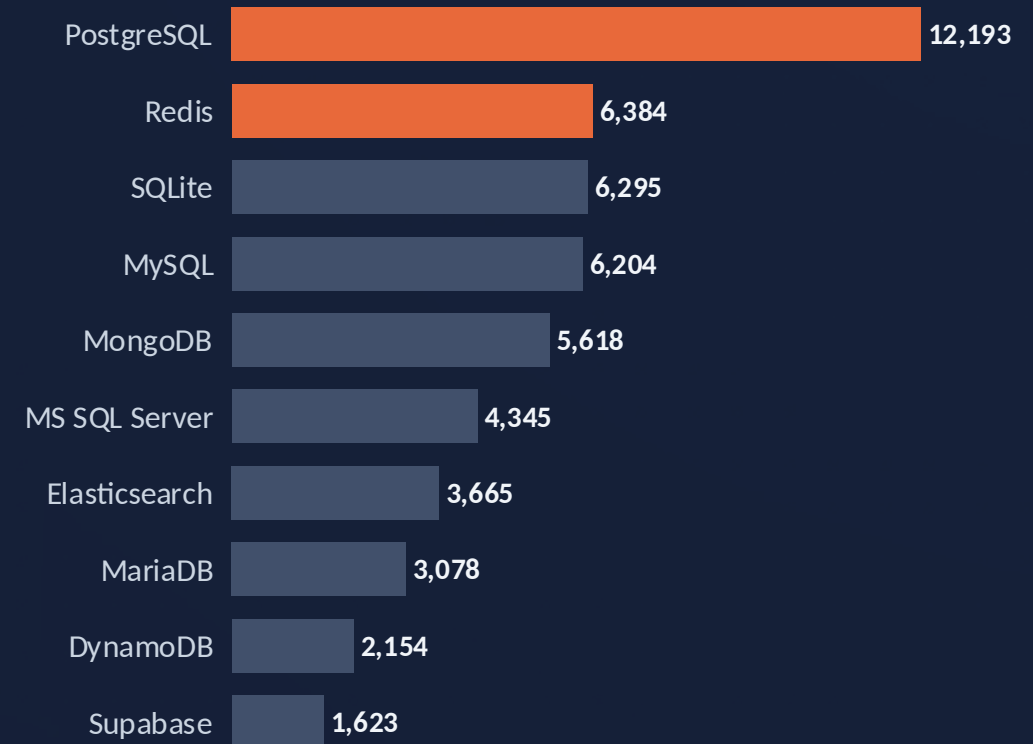
PostgreSQL widens its lead while Redis jumps to 2nd place

PostgreSQL leads both years with a growing gap; Redis jumps to 2nd place and Supabase enters next year's top 10.

Current year: databases used



Next year: databases desired



PostgreSQL is the database skill to prioritize

FINDINGS

- PostgreSQL is #1 in both years and grows from 11,514 to 12,193
- Redis rises from 6th to 2nd place (6,384)
- MySQL drops from 2nd to 4th (8,556 to 6,204)
- Supabase enters the top 10; Oracle leaves it

IMPLICATIONS

- PostgreSQL is the database skill to prioritize
- Caching and in-memory stores (Redis) are becoming more relevant
- Teams relying on MySQL or Oracle may need migration plans
- Managed, developer-friendly services (Supabase) are gaining interest

DASHBOARD

The interactive view: today, tomorrow and who answered

Google Looker Studio dashboard: Current Technology Usage, Future Technology
Trends and Demographics

02

Today's stack: JavaScript, PostgreSQL, AWS and Node.js/React lead



Key insights

- JavaScript is the most used language (14,943), followed by SQL, HTML/CSS, TypeScript and Python
- PostgreSQL leads the databases (11,514), ahead of MySQL and SQLite
- AWS is the dominant cloud platform (10,871); Microsoft Azure and Google Cloud follow
- Node.js (9,230) and React (8,999) far outpace every other web framework

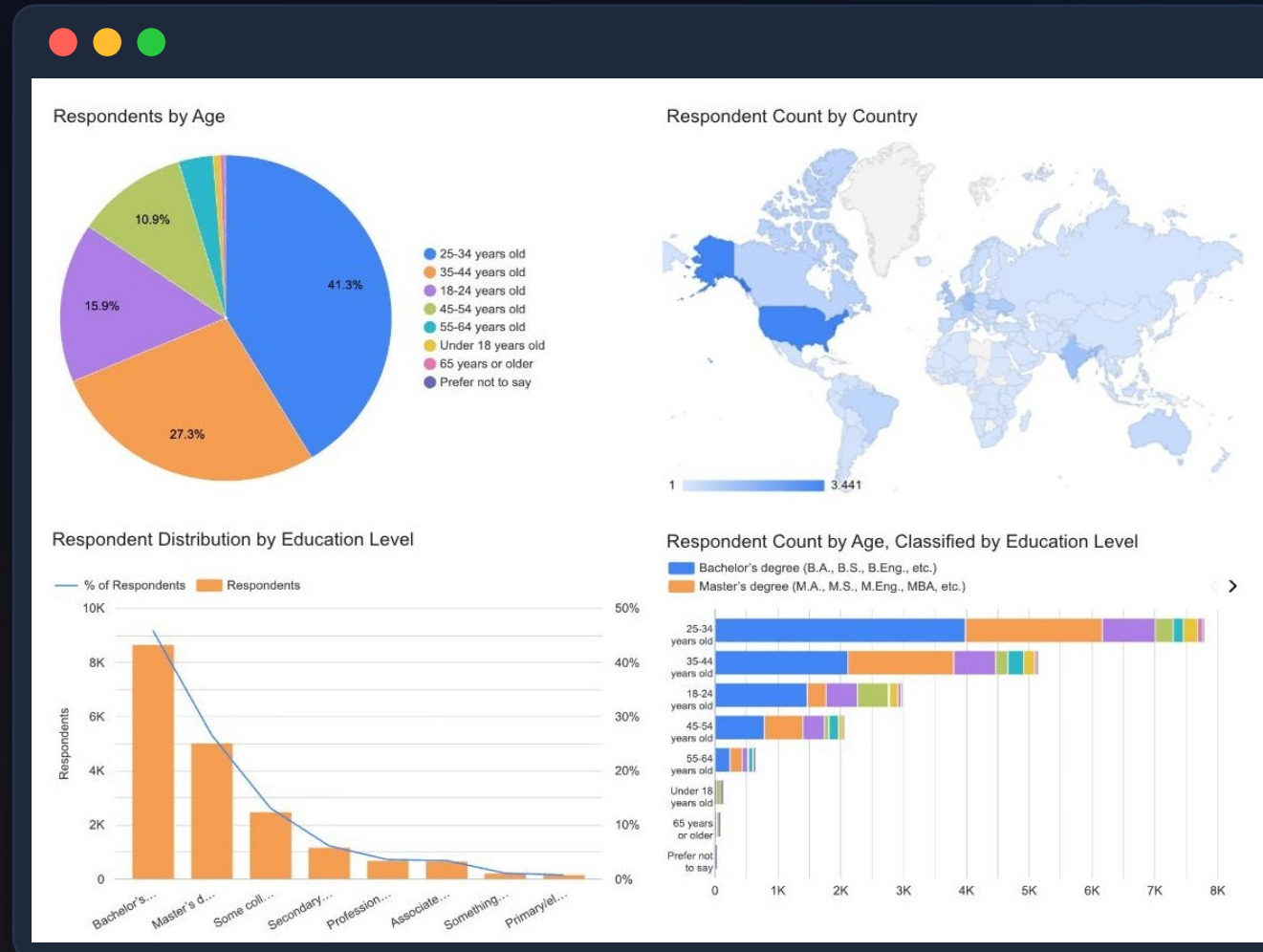
Tomorrow's stack: TypeScript, Rust, Go and Redis gain ground



Future trends

- JavaScript, SQL and TypeScript stay on top; TypeScript overtakes HTML/CSS
- Rust and Go enter the top 10, while PHP and PowerShell drop out
- PostgreSQL widens its lead; Redis rises to 2nd place and Supabase appears
- Cloudflare gains interest; Supabase replaces Heroku among platforms
- React and Node.js remain the most wanted; jQuery leaves the top 10, FastAPI and Svelte enter

Respondents are young, highly educated and mostly from the US



Demographic insights

- Respondents are young: 25-34 year olds are 41.3%, and 85% are under 45
- The United States has the most respondents (3,441), followed by Germany (1,339) and India (1,316)
- Bachelor's degree is the most common education level (45.8%), then Master's (26.5%)
- Bachelor's degrees lead in every adult age group; Master's degrees weigh more from age 35 (about 30%)

DISCUSSION & CONCLUSION

What it means for job seekers, companies and recruiters

Overall insights, findings & implications, and conclusion

03

The top of the stack is stable; change happens in the middle

Overall insights derived from the dashboards



Stable core

The technology stack is stable at the top: JavaScript, SQL, PostgreSQL, AWS and React/Node.js lead in both current usage and future interest



Risers

Change happens in the middle of the rankings: Rust, Go, Redis, Supabase, FastAPI and Svelte are gaining ground



Fading

Older technologies (PHP, PowerShell, Oracle, jQuery, Heroku) are losing interest



Caveat

The survey mainly reflects young, highly educated developers, mostly from the US and Europe, so results may not represent every market

Combine a solid core with one emerging skill

FINDINGS

Core skills are stable: JavaScript, SQL, Python, PostgreSQL and AWS stay at the top

Modern alternatives grow: TypeScript, Rust, Go, Redis and Next.js gain ground

Demand is concentrated geographically (Washington DC, Detroit, Seattle, New York)

Salary does not follow popularity: Swift pays the most, SQL and PHP the least



IMPLICATIONS

Job seekers should combine a solid core (SQL, JavaScript/Python) with one emerging skill

Companies should plan training on TypeScript, Rust/Go and cloud platforms

Recruiting efforts can focus on the cities with the most postings

Niche skills such as Swift can command a salary premium

Learn the stable core, watch the risers, follow the market

01



JavaScript, SQL and PostgreSQL are the most widely used technologies and remain the most desired

02



Rust, Go, TypeScript and Redis are the main growing technologies for next year

03



AWS, Azure and Google Cloud dominate cloud platforms; React and Node.js dominate web frameworks

04



Job opportunities and salaries vary strongly by location and language: monitoring them helps target skills and recruiting

Respondents come from 161 countries, led by the United States

Top 10 countries by number of survey respondents (161 countries in total)



3,441

respondents from the United States,
more than Germany (1,339) and India
(1,316) together

Median pay rises with age, peaking at about \$110K at 55-64

Median yearly compensation (USD) by age group



~\$25K → ~\$110K

Median pay rises steadily with age, from about \$25K (18-24) to a peak of about \$110K (55-64)

US developers earn 8.5x the median pay of developers in India

Median yearly compensation (USD) in the top 10 countries by respondents



\$65,000

Overall median yearly compensation (respondents reporting pay)

\$69,814

Median yearly compensation of full-time employees

$r = 0.41$

Correlation between compensation and work experience (outliers removed)

8.5x

Gap between US and India median compensation

Source: Stack Overflow Developer Survey, ConvertedCompYearly (respondents who reported compensation); exploratory analysis in Python (pandas, Seaborn)

Thank you for reading

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Notebooks, code and data sources are on GitHub:



github.com/v-ambrosino/ibm-data-analyst-capstone

